VXIII

FRAME CONSTRUCTION SPECIFICATION

STAGE 7 · HALITE AND FLEUR · REVISION 2.0 · OCTOBER 2026

VXIII

FRAME CONSTRUCTION SPECIFICATION

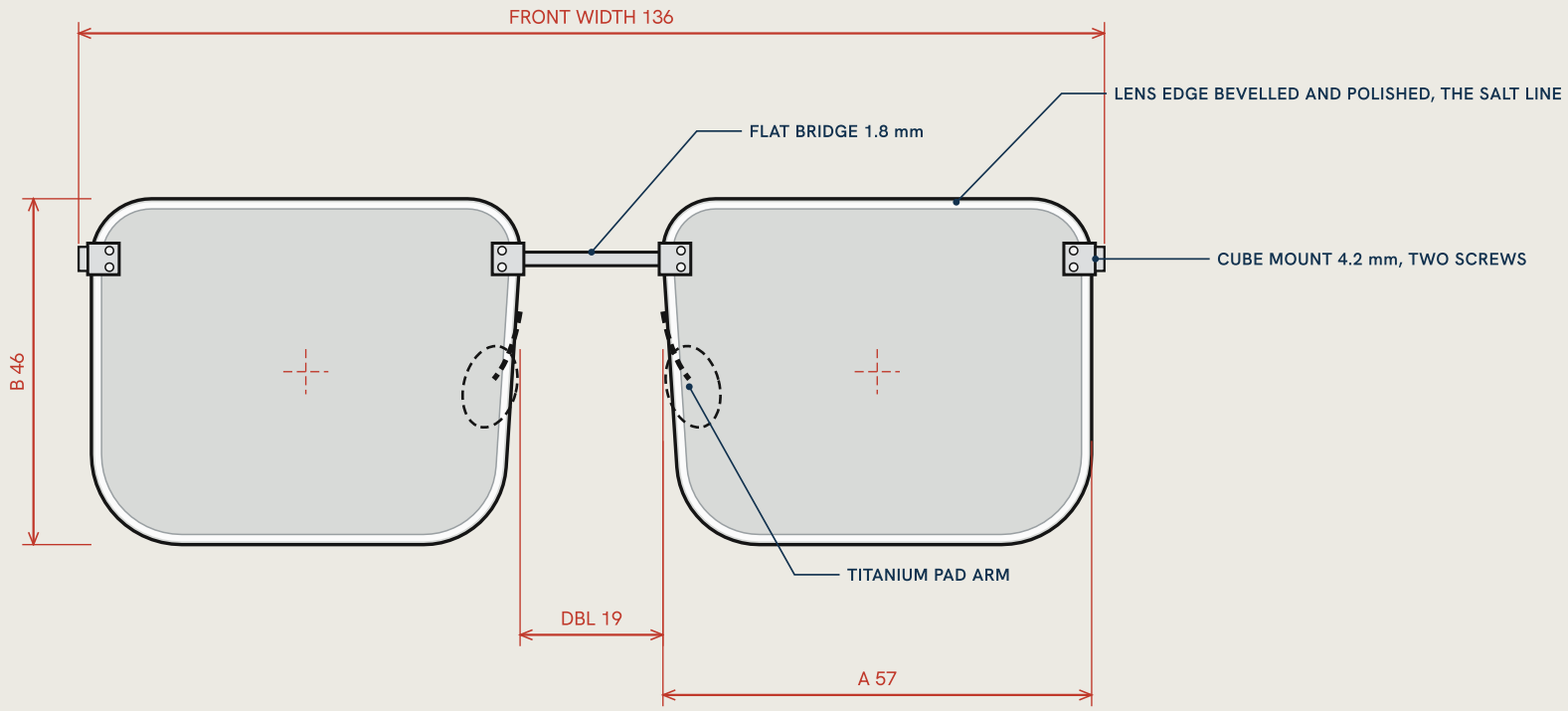
This document fixes every dimension, part, material, finish and tolerance needed to prototype and manufacture the two VXIII frames. It is drawn from the approved concepts H3 and F1 and the approved colourways.

All dimensions are in millimetres. Measurements follow the boxing system of ISO 8624: A is the width of the box round the lens, B its height and DBL the distance between the two boxes.

ITEM	HALITE AND FLEUR
Model	Halite, men · Fleur, women
Construction	Halite: rimless, three piece, drilled lens · Fleur: full rim cat eye in titanium wire
Box size	Halite 57 □ 19 145 · Fleur 56 □ 17 140
Lens height B	Halite 46.0 · Fleur 47.5
Front width	Halite 136 · Fleur 138
Colourways	Halite Lava, Brine, Smoke, Uyuni · Fleur Guérande, Alaea, Retba, Kala
Estimated weight	Halite about 16 g · Fleur about 18 g, confirmed at prototype

HALITE

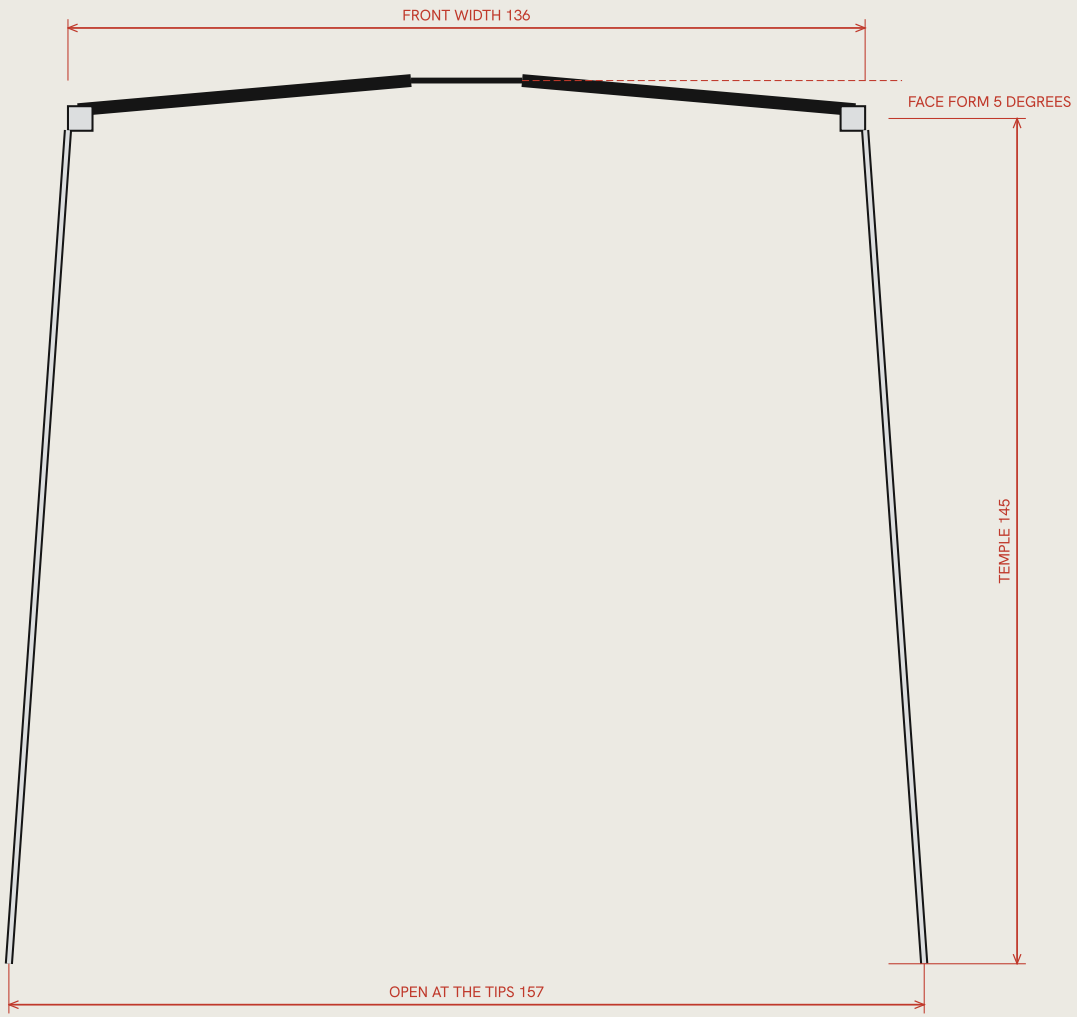
FRONT VIEW



FRONT VIEW AT FULL SIZE, 1:1. BOX SIZE 57 ▫ 19 145. LENS CENTRE DISTANCE 76 mm. DASHED LINES SHOW THE PAD ARMS BEHIND THE LENS.

HALITE

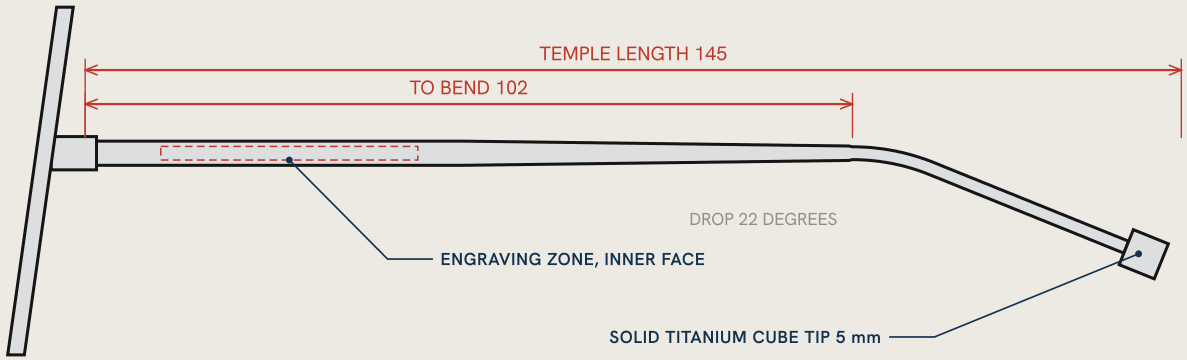
PLAN VIEW



PLAN VIEW AT 1:1.29. THE FRONT CARRIES A FACE FORM OF 5 DEGREES AND THE TEMPLES OPEN TO 94 DEGREES.

HALITE

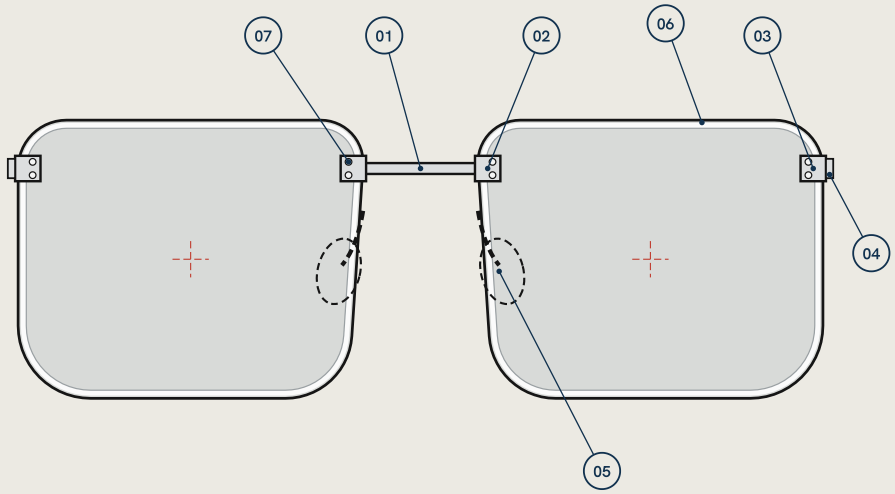
TEMPLE



ZONE, MM FROM THE HINGE	SECTION AND PURPOSE
Flat zone, 0 to 50	3.2 x 1.5, carries the engraving on the inner face
Taper, 50 to 101	3.2 falling to 1.9, the spring of the temple
Drop, 101 to 140	1.9 x 1.4, bent down 22 degrees behind the ear
Tip, 140 to 145	Solid titanium cube 5 mm, edges broken 0.4
Material	Beta titanium for spring and memory

HALITE

PARTS

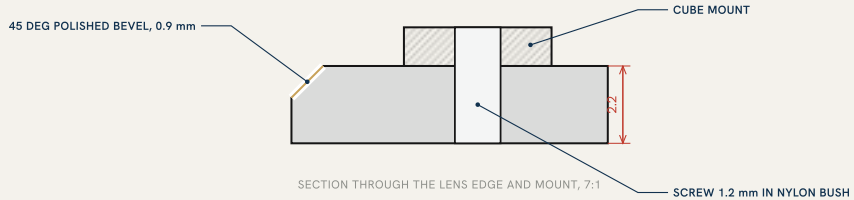


01 Bridge
02 Nasal cube mount
03 Temporal cube mount
04 Hinge
05 Pad arm and pad
06 Lens and salt line
07 Screws and bushes

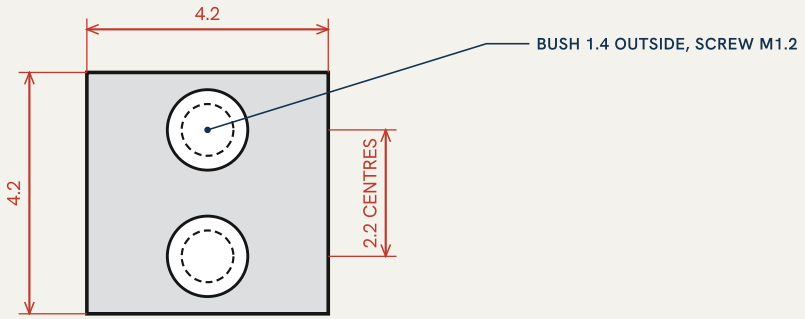
PART	NAME	MATERIAL AND FORM	QTY
H01	**Bridge**	CP titanium Grade 2 bar 1.8 x 1.2, welded between the nasal mounts	1
H02	**Nasal cube mount**	CP titanium Grade 2, 4.2 mm cube, two bores for bushes	2
H03	**Temporal cube mount**	CP titanium Grade 2, 4.2 mm cube with hinge barrels	2
H04	**Hinge**	Two barrel, barrels 2.4 mm, screw M1.2 x 4.0 in Grade 5 titanium	2
H05	**Pad arm and pad**	CP titanium wire 0.8, adjustable, silicone pad 11 x 7	2
H06	**Lens**	Polarised Trivex 2.2 mm, drilled four holes, edge bevel 0.9 x 45 degrees polished	2
H07	**Screws and bushes**	M1.2 titanium screws in nylon bushes 1.4 outside, four per lens	8
H08	**Temple**	Beta titanium, flat 3.2 x 1.5 to 50 mm, taper to 1.9 at the bend	2
H09	**Tip**	Solid titanium cube 5 mm, edges broken 0.4. Halite Smoke: acetate sleeve 34 mm	2

HALITE

LENS AND MOUNT DETAILS



SECTION THROUGH THE LENS EDGE AND MOUNT, 7:1

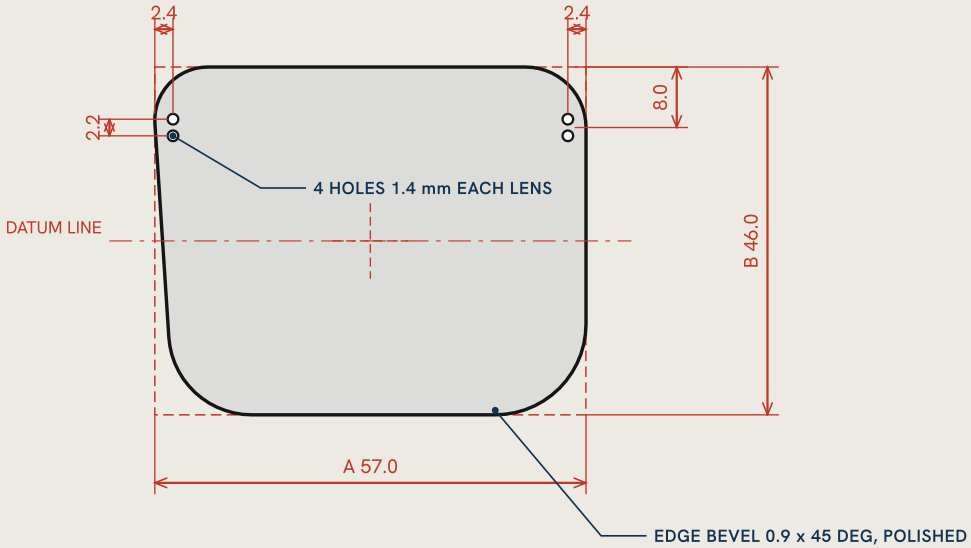


CUBE MOUNT, FRONT, 8:1

ITEM	SPECIFICATION
Lens edge	Bevelled 0.9 mm at 45 degrees on the front edge and polished clear, so the edge shows as a white line, the salt line
Mounting	Two holes per mount, 1.4 mm, at 2.2 mm centres, 2.4 mm from the lens edge. Nylon bushes, M1.2 titanium screws, thread lock
Bridge	Flat bar 1.8 x 1.2, laser welded between the two nasal mounts
Assembly torque	Screws tightened to the bush, never to the lens, so the lens is held without stress

HALITE

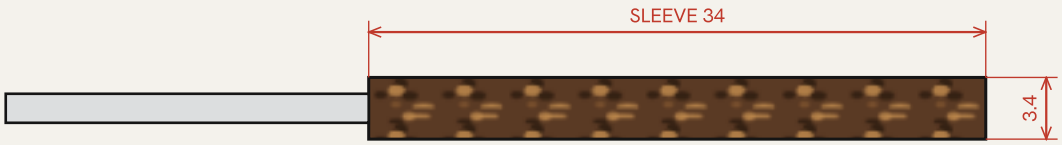
LENS TEMPLATE



ITEM	SPECIFICATION
Material	Polarised Trivex, 2.2 mm centre thickness, base 4 curve, chosen because it can be drilled without cracking
Size	A 57.0, B 46.0, drawn at 1:1 for the right lens and mirrored for the left
Filter category	Category 3 on all four colourways, 8 to 18 percent light transmission, UV400
Coatings	Hard coat and oleophobic outer coat. Uyuni adds a silver flash mirror on the outer face

HALITE

HALITE SMOKE TEMPLE SLEEVE

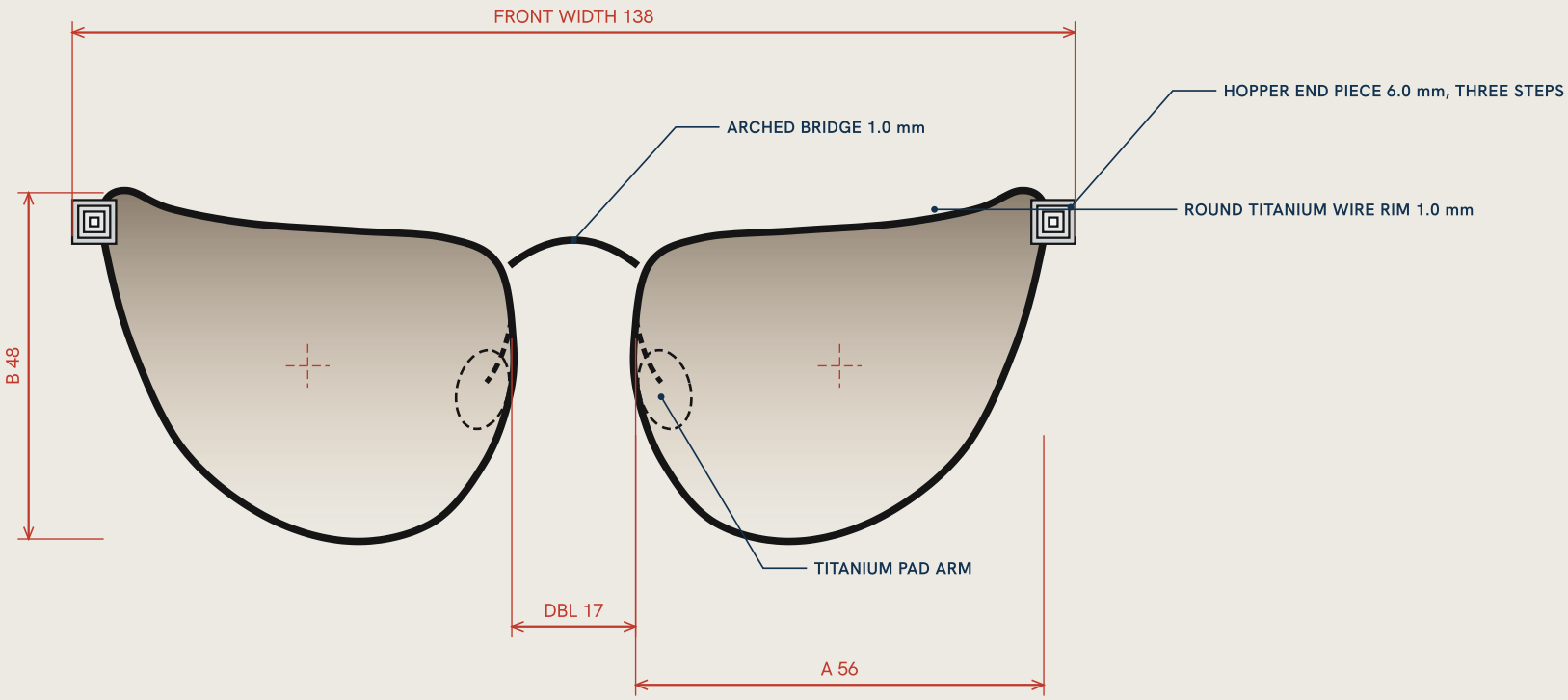


TITANIUM TEMPLE CORE RUNS INSIDE THE SLEEVE

ITEM	SPECIFICATION
Sleeve	Cellulose acetate in tortoise, 34 mm long, 3.4 mm high, pushed over the titanium core of the temple and heat set
Replaces	The solid titanium cube tip on Halite Smoke only. The other three Halite colourways keep the cube tip
End	Square end, edges broken 0.4, so the sleeve finishes in the same square as the cube

FLEUR

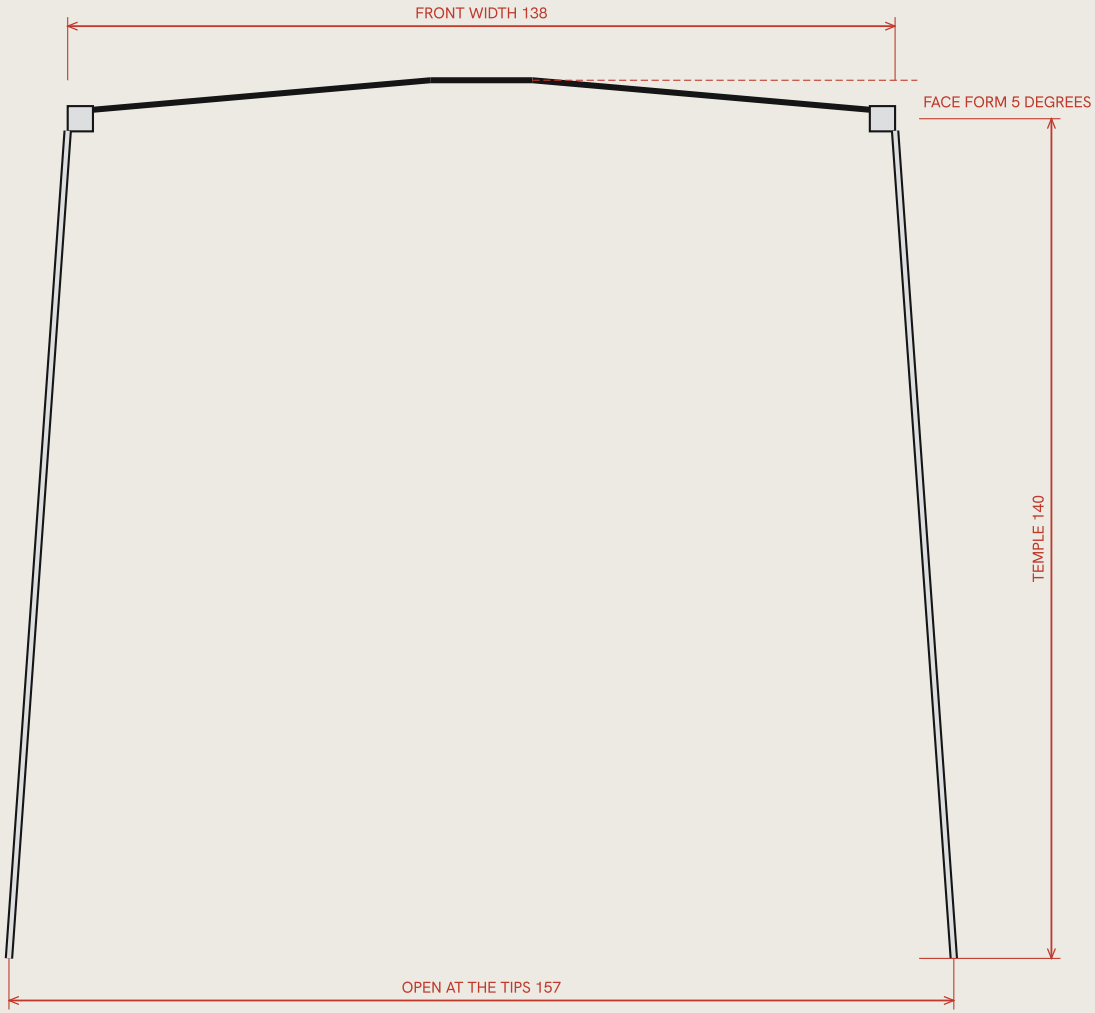
FRONT VIEW



FRONT VIEW AT FULL SIZE, 1:1. BOX SIZE 56 ▫ 17 140. LENS CENTRE DISTANCE 73 mm. THE OUTER CORNER IS LIFTED 4 mm ABOVE THE TOP LINE.

FLEUR

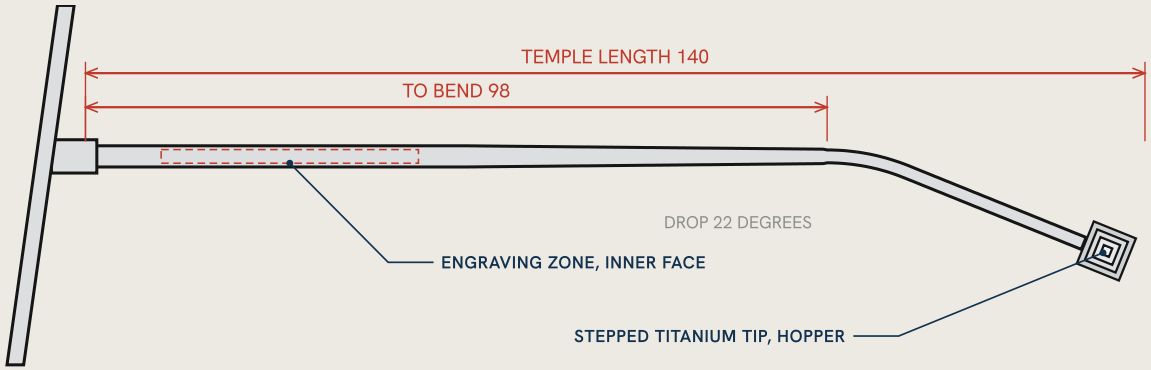
PLAN VIEW



PLAN VIEW AT 1:1.26. THE FRONT CARRIES A FACE FORM OF 5 DEGREES AND THE TEMPLES OPEN TO 94 DEGREES.

FLEUR

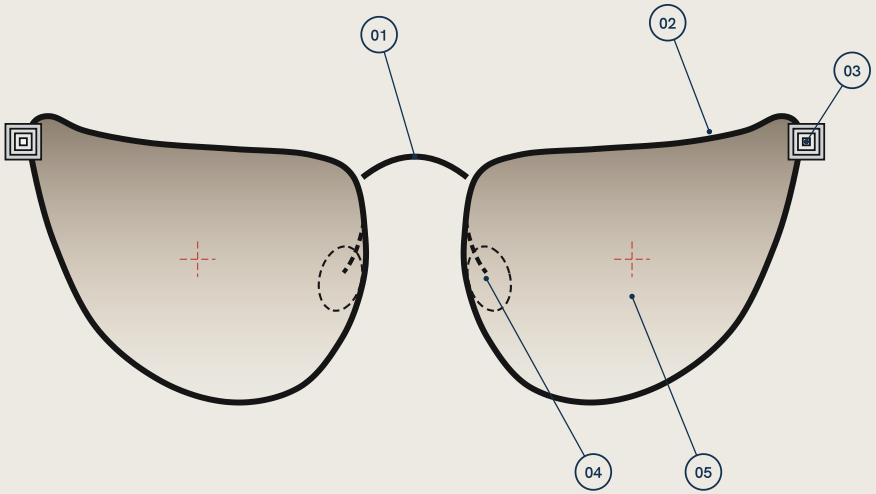
TEMPLE



ZONE, MM FROM THE HINGE	SECTION AND PURPOSE
Flat zone, 0 to 50	2.8 x 1.4, carries the engraving on the inner face
Taper, 50 to 98	2.8 falling to 1.9
Drop, 98 to 134	Round 1.4, bent down 22 degrees behind the ear
Tip, 134 to 140	Stepped titanium square, the hopper repeated
Material	Beta titanium for spring and memory

FLEUR

PARTS

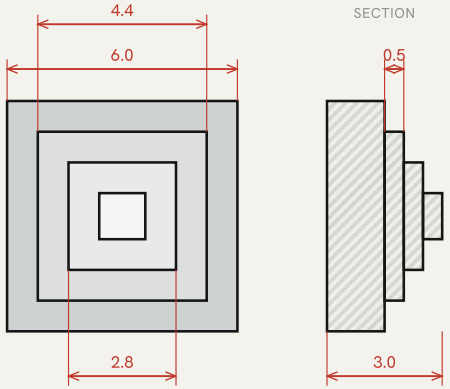


01 Bridge
02 Rim wire
03 Hopper end piece and hinge
04 Pad arm and pad
05 Lens

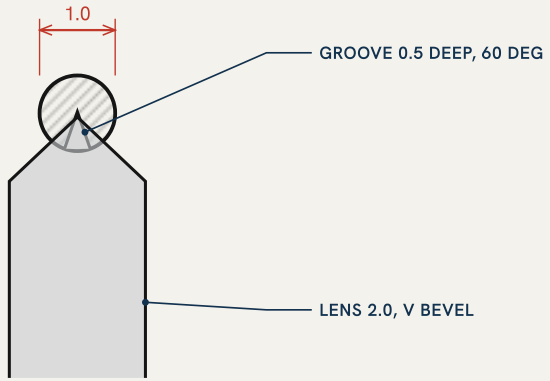
PART	NAME	MATERIAL AND FORM	QTY
F01	**Bridge**	CP titanium wire 1.0, arched, welded to the rims	1
F02	**Rim**	CP titanium wire 1.0, grooved 0.5 for the lens, closed at the end piece	2
F03	**Hopper end piece**	CP titanium Grade 2, squares of 6.0, 4.4, 2.8 and 1.2 stepped 0.5, hinge behind	2
F04	**Hinge**	Two barrel, barrels 2.4 mm, screw M1.2 x 4.0 in Grade 5 titanium	2
F05	**Pad arm and pad**	CP titanium wire 0.8, adjustable, silicone pad 11 x 7	2
F06	**Lens**	Polarised polyamide 2.0 mm, gradient, V bevel	2
F07	**Temple**	Beta titanium 1.4 wire, flattened to 2.8 x 1.4 for 50 mm	2
F08	**Tip**	Stepped titanium square, the hopper repeated. Fleur Alaea: acetate sleeve 34 mm	2

FLEUR

HOPPER END PIECE AND RIM



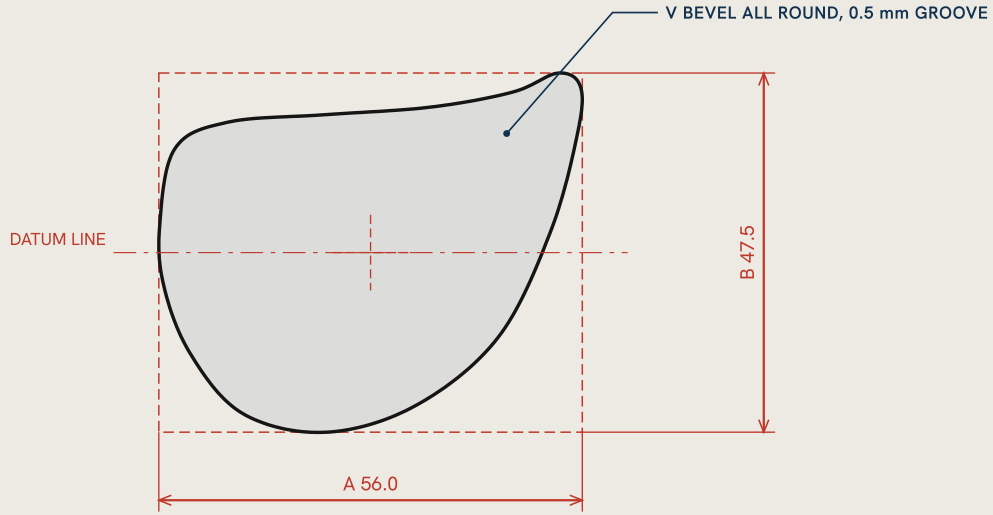
HOPPER END PIECE, FRONT AND SECTION, 6:1



RIM WIRE SECTION, 10:1

ITEM	SPECIFICATION
Hopper end piece	Machined from a single block of CP titanium. Four squares of 6.0, 4.4, 2.8 and 1.2 mm, each step 0.5 mm deep, total depth 3.0 mm. Edges kept sharp and polished
Rim	Round CP titanium wire 1.0 mm with a 0.5 mm groove at 60 degrees, closed at the end piece by a laser weld hidden behind the hopper
Bridge	Round wire 1.0 mm, arched, rising 7 mm at its centre

FLEUR LENS TEMPLATE



ITEM	SPECIFICATION
Material	Polarised polyamide, 2.0 mm centre thickness, base 6 curve, gradient dark at the top
Size	A 56.0, B 47.5, drawn at 1:1 for the right lens and mirrored for the left
Filter category	Measured at the reference point: category 3 for Guérande, Alaea and Kala, category 2 for Retba. UV400
Edge	V bevel all round to sit in the 0.5 mm rim groove

FLEUR ALAEA TEMPLE SLEEVE



TITANIUM TEMPLE CORE RUNS INSIDE THE SLEEVE

END FACE, STEPPED SQUARE
ENGRAVED 0.2 DEEP

ITEM	SPECIFICATION
Sleeve	Cellulose acetate in red tortoise, 34 mm long, 3.4 mm high, pushed over the titanium core and heat set
Replaces	The stepped titanium tip on Fleur Alaea only. The other three Fleur colourways keep the titanium hopper tip
End face	The stepped square of 3.4, 2.4 and 1.4 mm engraved 0.2 mm deep into the square end of the sleeve, so the hopper stays on every Fleur

BOTH

ENGRAVING

LEFT TEMPLE, INNER FACE, HALITE



ENGRAVING ZONE FROM 10 mm, WORDMARK STARTS AT 12.4 mm

RIGHT TEMPLE, INNER FACE, HALITE



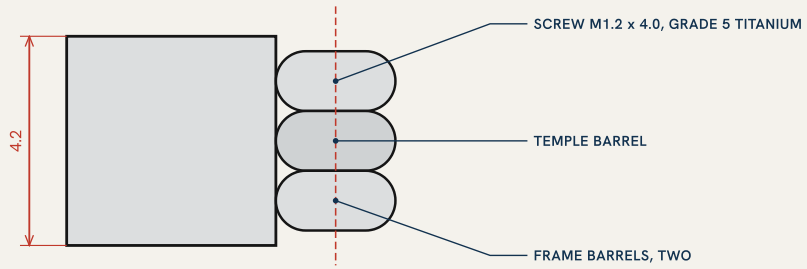
ENGRAVING ZONE FROM 10 mm, WORDMARK STARTS AT 12.4 mm

ITEM	SPECIFICATION
Left temple	The VXIII wordmark, the verse line 2.2 mm after it, then the size mark apart at 44 mm, as ISO 12870 requires
Right temple	KEEP YOUR SAVOUR in capitals, alone
Size	Wordmark cap height 2.0 mm on Halite and 1.8 mm on Fleur. Verse line and size mark 0.5 mm. Tagline 0.7 mm
Method	Laser through the finish to 0.14 mm, no filler, so the cut shows bare titanium
Not marked	Lens, front of the frame and outer faces of the temples

STRIPS DRAWN AT ABOUT 3:1. FLEUR USES THE SAME LAYOUT ON ITS 2.8 mm TEMPLE.

BOTH

HINGE AND NOSE PADS



TWO BARREL HINGE, PLAN, 8:1

ITEM	SPECIFICATION
Hinge	Two barrel, barrels 2.4 mm diameter, frame barrels in the mount or end piece, temple barrel between them
Screw	M1.2 x 4.0, Grade 5 titanium, thread lock on assembly
Opening	Temples open to 94 degrees to the front and stop on the mount face
Pad arm	CP titanium wire 0.8, welded to the bridge side of the mount or rim, adjustable by the optician
Pad	Medical grade silicone, clear, 11 x 7 mm, push fit

BOTH

MATERIALS AND FINISHES

COLOURWAY	FINISH	METHOD	LENS	MASTER
Halite Lava	Matte black titanium	PVD coating	Grey polarised	#1F1F1F
Halite Brine	Polished gold titanium	PVD coating	Green polarised	#C8A25A
Halite Smoke	Brushed bronze titanium with tortoise acetate temple ends	PVD coating, acetate sleeves	Brown polarised	#6E4E36
Halite Uyuni	Brushed gunmetal titanium	PVD coating	Grey polarised with a silver mirror	#53575A
Fleur Guérande	Polished gold titanium	PVD coating	Brown gradient polarised	#C9A55E
Fleur Alaea	Rose gold titanium with red tortoise acetate temple ends	PVD coating, acetate sleeves	Copper gradient polarised	#C38F78
Fleur Retba	Champagne titanium	PVD coating	Rose gradient polarised	#D9C3A1
Fleur Kala	Matte black titanium	PVD coating	Grey gradient polarised	#1F1F1F

ITEM	SPECIFICATION
Titanium	CP titanium Grade 2 for mounts, end pieces and bridge. CP titanium Grade 1 wire for the Fleur rim. Beta titanium for temples. Grade 5 for screws
Acetate	Cellulose acetate sleeves for Halite Smoke and Fleur Alaea
Colour	PVD coating at 0.5 to 1.0 micron after polishing or brushing. Colour matched to the sealed master sample

BOTH

FIT

PARAMETER	HALITE	FLEUR	NOTE
Box size	57 □ 19 145	56 □ 17 140	Engraved small inside the left temple, after the verse
Front width	136	138	Suits a face width of about 132 to 145 mm
Lens centre distance	76	73	Close to the common adult pupillary distance
Face form	5 degrees	5 degrees	Follows the curve of the face
Pantoscopic tilt	8 degrees	8 degrees	Front leans toward the cheek
Temple length	145	140	Drop of 22 degrees behind the ear
Nose pads	Adjustable	Adjustable	Fit and comfort is the first reason to buy, at 82 percent

BOTH

MANUFACTURE

STEP	OPERATION	CONTROL
01	Machine the cube mounts, the hopper end pieces and the cube tips	Dimensions against the drawing, edges sharp
02	Form the Fleur rim wire and bridge, and the temples	Template check at 1:1
03	Laser weld bridges, end pieces, pad arms and hinge barrels	Weld visual and pull test
04	Polish or brush, then clean	Surface against the master sample
05	PVD colour	Colour against the master, thickness record
06	Engrave the inner faces of the temples, left and right	Depth 0.14 mm, zone from 10 mm
07	Cut, drill and bevel the Halite lenses, cut and V bevel the Fleur lenses	Hole position within 0.05 mm, bevel polished clear
08	Assemble lenses, hinges, pads and sleeves	Lens held without stress, hinge swing test
09	Final inspection and packing	Dimensional and cosmetic check, UV test per batch

BOTH

TOLERANCES AND TESTING

DIMENSION	TOLERANCE
Lens width A	± 0.2
Lens height B	± 0.3
Bridge DBL	± 0.3
Temple length	± 1.0
Front width	± 0.5
Drill hole position, Halite	± 0.05
Hopper step depth, Fleur	± 0.03
Engraving depth	± 0.03
Face form and pantoscopic tilt	± 1 degree

TEST	REQUIREMENT
Sunglass lenses	ISO 12312-1 for the filter category of each colourway
Frames	ISO 12870 for dimensions and robustness
Rimless mounting	No lens movement or cracking after the robustness test
Corrosion	Artificial perspiration test, no visible change to finish
Nickel release	Within the limit of each market
Finish adhesion	No flaking after the tape and abrasion test

THE MASTER SAMPLE OF EACH COLOURWAY IS THE REFERENCE FOR COLOUR AND SURFACE AND IS KEPT SEALED.